

Launchpad-AI

The Launchpad AI Appliance is a bootable ISO that you install on bare metal to create a GPU-enabled Kubernetes cluster

preconfigured for AI workloads. The ISO is built with an immutable [Kairos](#)-based operating system, Palette eXtended Kubernetes Edge (PXK-E), the NVIDIA GPU Operator, replicated block storage, and the Launchpad AI platform.

The overall installation flow is:

1. Download the Launchpad AI ISO from Artifact Studio.
2. Flash the ISO to bootable media and boot each bare metal node from it.
3. Use the **Palette Edge Interactive Installer** to install the OS onto the local disk.
4. Use the **Palette TUI** to configure the node's IP address and network settings.
5. Use **Local UI** to link the hosts and deploy the cluster.

The installation procedure is identical to the [Palette Management Appliance](#).

Hardware Requirements

Each device where you install the Launchpad AI Appliance ISO must meet the following requirements.

Component	Minimum	Additional Information
CPU	Intel or AMD64 CPU with 8 cores	-
GPU	One or more NVIDIA GPUs	Required for AI workloads. The NVIDIA GPU Operator installs the drivers automatically.
RAM	24 GB	Increase based on the size and number of models you intend to run.
Network Adapters	2 x 1 Gbps (data and management)	A dedicated, high-bandwidth interface is recommended for storage replication traffic.
Disks	Local disk of at least 500 GB for the OS boot, plus a second disk for the storage pool	The storage pool drive must be separate from the OS boot disk.

Prerequisites

- Download the [Launchpad AI ISO](#) and the [content bundle](#) (`content.zst`).
- A second, dedicated storage device on each node for the Piraeus storage pool (for example, `/dev/sdb`).
- A reserved virtual IP address (VIP) for the cluster, and an unused IP address (or range) for MetalLB to expose platform services.

Install the OS with the Interactive Installer

1. Download the [Launchpad AI ISO](#) and the [content bundle](#) (`content.zst`).
2. Flash the ISO to a bootable device, such as a USB stick. You can use tools such as [balenaEtcher](#), or transfer the ISO to the node with `scp` or `rsync` .
3. Attach the bootable media to each bare metal node and set the boot order to boot from the Launchpad AI ISO first.
4. Restart the node to start the installation process.
5. When the node reaches the GRand Unified Bootloader (GRUB) menu, allow it to select the **Palette Edge Interactive Installer** boot option automatically.
6. When the **Palette Edge Interactive Installer** window appears, select the target disk to install the OS onto.
Ensure you select the correct disk. The installation process erases all content on the target disk. Do not select the disk you intend to use as the storage pool drive.
7. Choose the post-installation action — **nothing**, **reboot**, or **poweroff** — then press **ENTER**.

Option	Description
nothing	Keeps the system powered on after installation.
reboot	Automatically reboots the system.
poweroff	Powers off the system.

8. Review the **Installation Summary** and press **ENTER** to start the installation.
9. Wait for the installation to complete. This takes at least 15 minutes, depending on the node's resources. After it completes, disconnect the ISO. If you selected **reboot**, the node reboots and displays the Palette TUI. If you selected **poweroff**, power the node back on.

Configure the IP Address with the Palette TUI

1. In the **Palette TUI**, provide credentials for the initial account. Use this account to log in to Local UI and to access the node over SSH.

Field	Description
Username	Provide a username to use for the account.
Password	Enter a password for the account.
Confirm Password	Re-enter the password for confirmation.

Press **ENTER** to continue.

2. The available configuration options appear. Use the **TAB** key or the up and down arrow keys to switch between fields. When you make a change, press **ENTER** to apply it. Use **ESC** to go back.
3. In **Hostname**, check the existing hostname and, optionally, change it.
4. In **Network Adapter**, select the adapter to configure. By default, adapters request an IP address automatically from the DHCP server. For each adapter, you can:
 - Switch from DHCP to **static IP**, and provide a static IP address, subnet mask, and default gateway. Setting a static IP removes the existing DHCP settings.
 - Specify a **VLAN ID** to segment traffic on the same physical interface.
 - Specify the **MTU**, the largest packet size the interface can send without fragmentation.
5. In **DNS Configuration**, specify the primary and alternate name servers, and optionally a search domain.
6. In **NTP Configuration**, specify one or more NTP servers. For example, `0.pool.ntp.org` .
7. After you are satisfied with the configuration, navigate to **Quit** and press **ENTER**. Press **ENTER** again on the confirmation prompt. After a few seconds, the terminal displays the **Device Info** and the Local UI address.

To access the Palette TUI again, run the `palette-tui` command in the terminal.
8. Repeat the OS installation and Palette TUI configuration on every node before continuing.

Link the Hosts in Local UI (skip for single node installation)

1. Decide which host is the leader of the group. The leader coordinates linking the other hosts. For multi-node clusters, use an odd number of nodes in the control plane for high availability.
2. In a web browser, go to `https://<node-ip>:5080` . Replace `<node-ip>` with the leader host's IP address. The Local UI address also appears on the node's terminal screen. If you changed the default port, replace `5080` with your configured port.

Local UI uses a self-signed certificate, so your browser may show a security warning the first time. You can safely proceed.
3. Log in to Local UI with the credentials you created in the Palette TUI.
4. From the left main menu, select **Linked Edge Hosts**.
5. Select **Generate token**. The leader generates a Base64-encoded token that contains its IP address and a one-time password (OTP) that expires after two minutes. Select **Copy** to copy the token.
6. Log in to Local UI on each host you want to link, select **Linked Edge Hosts**, then select **Link this device to another**. Enter the token you copied from the leader and select **Confirm**.
7. Repeat for every host you want to link to the leader.
8. Confirm that all linked hosts appear in the **Linked Edge Hosts** table with the following statuses.

Column	Status
Status	Ready
Content	Synced
Health	Healthy

Content synchronization takes at least five minutes, depending on your network.

Deploy the Cluster

1. From the left main menu, select **Content**, then click on **Actions** and **Upload Content**.
2. Wait for content to be uploaded successfully.
3. From the left main menu, select **Cluster**, then select **Create cluster**.
4. Complete the **Basic Information** fields and select **Next**.

Parameter	Description
Cluster name	Name of the cluster.
Tags	Key-value pairs to provide metadata about your cluster.

5. In **Cluster Profile**, the default Launchpad AI profile loads with the packs described in the [Architecture](#) section. Review the profile and select **Next**.
6. In **Profile Config**, configure the profile variables for the cluster. Several variables ship with defaults that you can keep or adjust. For the full list of variables, their descriptions, types, and defaults, see [Profile Variables](#) at the end of this guide. Select **Next** when you are done.
7. In **Cluster Config**, configure the following options, then select **Next**.

Parameter	Description	Default
Virtual IP Address (VIP)	The virtual IP address for the cluster, used for load balancing and HA.	<i>No default</i>
Network Time Protocol (NTP)	(Optional) The IP address of an NTP server the cluster can reference.	<i>No default</i>
SSH Keys	(Optional) Public SSH keys to access the cluster nodes.	<i>No default</i>

8. In **Node Config**, assign hosts to the control plane and worker node pools. To assign a host, select **Add Item** in the node pool and select the host. Keep the leader node in the control plane pool, and use an odd number of control plane nodes. You can remove the worker pool if it is not required, but ensure **Allow worker capability** is enabled on the control plane pool when you do. Select **Next**.
9. Review your configuration and deploy the cluster. The **Cluster** page displays the deployment status. The nodes reboot as part of the build process, and deployment can take up to 45 minutes.

Validate

1. On the **Cluster** page, confirm that the cluster reaches a **Running / Healthy** state and that all packs install successfully.

If the installation stalls, verify that the `piraeus-operator` and `nvidia-gpu-operator-ai` packs installed

correctly. GPU driver installation can take additional time on first boot.

2. After deployment completes, additional options appear in the left main menu. Use them to access the Launchpad AI platform and confirm that GPU nodes are schedulable.

Next Steps

After you deploy your Launchpad AI cluster, configure the Launchpad AI platform and deploy your first AI workload.

Profile Variables

The Launchpad AI cluster profile exposes the following profile variables. You set these in the **Profile Config** step

when you [deploy the cluster](#). The **Variable** column shows the template reference used in the pack values (for example, `{{.spectro.var.k8s.podCidr}}`).

Display Name	Variable	Used By	Description	Type	Default
Pod CIDR	<code>k8s.podCidr</code>	<code>edge-k8s</code>	The CIDR range for the pod network. Used to allocate IP addresses to pods. Change only if it conflicts with your network.	CIDR notation	<code>100.64.0.0/18</code>
Service CIDR	<code>k8s.serviceCidr</code>	<code>edge-k8s</code>	The CIDR range for the service network. Used to allocate IP addresses to services. Must not overlap with the Pod CIDR.	CIDR notation	<code>100.64.64.0/18</code>
System Reserved	<code>os.systemReservedCPU</code>	<code>edge-native-byoi</code>	CPU core IDs reserved for	String	<i>No default</i>

CPU			the OS and the Kubernetes node agent (kubelet). These cores are excluded from workloads. Example: 0-3 .		
MetalLB IP Range	metallb.ipRange	lb-metallb	The IP address (or range) that MetalLB uses to expose platform services externally. Use an unused address on your network.	String	<i>No default</i>
L2 Interface	l2interface	lb-metallb	The physical network interface, bond, or bridge used by MetalLB for L2 advertisement of the platform IP.	String	<i>No default</i>
Storage Pool Drive	csi.storagepoolDrive	piraeus-operator	The storage pool device to use for the cluster. Assign this to your second (dedicated) storage device, not	String	/dev/sdb

			the OS boot disk.		
Storage Node Interface	csi.storageNodeInterface	piraeus-operator	The network interface on each node dedicated to storage (DRBD) replication traffic. Choose a high-bandwidth interface or bond.	String	No default
CSI Placement Count	csi.placementCount	piraeus-operator	The number of replicas maintained for each persistent volume. Accepted values are 1 or 3. Use 3 for high availability.	Integer	3
Launchpad AI Username	launchpad_ai_username	launchpad-ai	The username used to authenticate to the Launchpad AI platform.	String	No default
Launchpad AI Password	launchpad_ai_password	launchpad-ai	The password used to authenticate to the Launchpad AI platform.	String	No default